

Human Natural Structure

Foundational & Observational Specification

*The Canonical Coordinate System for Structural and Observational
Consistency*

Volume 2 · Human Natural Structure Series

Satoru Hara

Natural Structure Works (NSW)

A Note on Method and Evidence

This volume specifies the Axis-2 coordinate of Human Natural Structure (HNS): the 6×6 base matrix HNS-36 and its observational extension HNS-144. It is a structural specification, not an empirical report, and it keeps three registers distinct throughout.

- **(i) Established results** of the cited literatures — cognitive control, dual-process theory, upper ontologies, causal inference — are reported as such.
- **(ii) The HNS design programme** — the engineering proposal that a fixed Layer×Category coordinate can screen structural failures in language-model output. Most of this book belongs here.
- **(iii) Open conjectures** — that the chosen partition is complete, and that its cells correspond to identifiable brain functions. These are flagged where they occur.

Two honesty commitments carry over from the Origin Trilogy. First, the lineage attributions in this book indicate conceptual inspiration, not strict derivation from any single source ontology. Second, and importantly, the recurring number six is a chosen resolution, not a discovered natural constant: no source ontology has exactly six categories, and the layer stack is a deliberate partition. The framework is more credible for owning this than for implying metaphysical necessity. A specification candid about its evidentiary boundaries is harder to dismiss than one that is not.

Contents

- A Note on Method and Evidence2
- Contents.....3
- Preface — From a Verification Hypothesis to a Coordinate System.....5
- Part I.....7
 - 1. HNS-36 as the Axis-2 Cellular Matrix8
 - 1.1 Reading HNS-368
 - 1.2 Notation and conventions.....9
 - 2. The Two Axes.....10
 - 2.1 Why two independent axes multiply10
 - 3. Human Natural Layers (L1–L6)11
 - 3.1 Layer structure requirements.....11
 - 3.2 Boundary conditions and misclassification12
 - 3.3 The six layers in detail.....12
 - 4. Abstract Categories (C1–C6)15
 - 4.1 Category structure requirements.....15
 - 4.2 The six categories in detail.....16
 - 5. The 36-Cell Specification18
 - 5.1 Cell format18
 - 5.2 Structural and matrix constraints18
 - 5.3 Adjacency and interpretation.....18
 - 5.4 Mapping rules19
 - 5.5 Representative cell entries.....19
 - 5.6 The thirty-six cells at a glance.....25
 - 6. Layer Jump26
- Part II.....28
 - 7. HNS-144: The Observational Extension.....29
 - 7.1 The four quadrants in detail29
 - 8. Category Ambiguity.....31
 - 9. Mapping and Usage at 144 Resolution.....32
- Part III.....34
 - 10. Relationship to the HNS Architecture.....35
 - 10.1 The nested coordinate (Axis 2).....35
 - 10.2 The orthogonal grounding axis (Axis 3)35
 - 10.3 The runtime.....35
 - 10.4 Relationship to the three MAV axes.....35
 - 11. Intellectual Lineage and Proponents37
 - 11.1 The layers.....37
 - 11.2 The categories37
 - 11.3 The quadrants37
 - 11.4 Lineage at a glance38
 - 12. Empirical Status, Limitations, Threats to Validity.....39

12.1 What is established	39
12.2 What is proposed.....	39
12.3 What is conjecture.....	39
12.4 On the number six	39
12.5 Threats to validity	40
Appendix A (Normative) — HNS Unified Structural Coordinate Table	
v2.0.....	41
A.1 Natural Layers (HNS-36)	41
A.2 Abstract Categories (HNS-36)	41
A.3 Onto-Modal Quadrants (HNS-144).....	41
A.4 Inference & Control Operators (HNS-864)	41
A.5 Social Meta Layers (HNS-SMS-6) — parallel grounding axis (Axis 3).....	42
Appendix B — The 36-Cell Template (Formal)	43
Appendix C — Worked Examples.....	44
C.1 Layer Jump (HNS-36)	44
C.2 Category Ambiguity (HNS-144)	44
Glossary.....	45
References.....	47

Preface — From a Verification Hypothesis to a Coordinate System

The Origin Trilogy advanced a hypothesis: reliable cognition arises not from a single computational axis but from the geometric intersection of several approximately orthogonal verification processes — the Multi-Axis Verification (MAV) account. Of its three axes, Axis 2 (Structural / Causal) supplies constraints on permissible structure, category, and causal order, independent of statistical frequency. In a deployed language model, Axis 2 is realized as the HNS Cellular Matrix.

That trilogy described the matrix; it did not fully specify it. This volume does. Its task is narrow and foundational: to fix, once and explicitly, the coordinate system on which every later HNS specification depends. HNS-36 is the base coordinate — the Structural layer; HNS-144 is the finer observational resolution — the Observational layer. Together they are the Foundational and Observational specifications of Axis 2.

Why a coordinate system at all? Because description alone cannot say where a proposition sits. The Origin Trilogy and its companion volumes give a rich description of human, cognitive, and social structure, but description provides depth without position. A coordinate system supplies position: it lets an external verifier state precisely which layer a claim originates in and which category explains it, and therefore lets every correction be attributed to an explicit cell. The move is from terrain map to geodetic grid — from knowing the landscape to being able to give coordinates within it. Auditability, the property the programme can most securely demonstrate, is a direct consequence of having such coordinates.

What this book is. A normative specification of HNS-36 and HNS-144: the two axes, their cells, the rules for placing a proposition, and the two failure classes the coordinate is designed to screen — Layer Jump and Category Ambiguity. What this book is not. It is not a method, an evaluation model, or domain guidance, and it makes no claim that the coordinate is the only possible one. It defines a stable reference frame; how that frame is used is the subject of later volumes.

A word on how to read what follows. Part I specifies the foundational coordinate, HNS-36: its two axes, its cells, and the failure it screens. Part II adds the observational resolution, HNS-144. Part III places the result within the larger architecture and states, candidly, what has and has not

been shown. A reader who wants the normative core alone may read Chapters 3 through 5 with Appendix A; the surrounding chapters supply motivation, lineage, and limitation.

Part I
The Foundational Coordinate — HNS-36

1. HNS-36 as the Axis-2 Cellular Matrix

HNS-36 is a 6×6 coordinate system formed by the intersection of six Natural Layers (the causal axis) and six Abstract Categories (the explanatory axis). Each of its thirty-six cells is a structural position — the meeting of one layer and one category — not a bin into which phenomena are sorted by type. This distinction is foundational: HNS-36 is a coordinate system, not a taxonomy. It positions a proposition by causal origin × explanatory purpose; it does not classify it by kind.

To see that this is a coordinate and not a taxonomy, consider a single phenomenon — a company's falling share price. A taxonomy would ask what kind of thing it is and file it once. The matrix instead asks two coordinate questions and admits several legitimate placements depending on interpretive purpose: as a macro-social state ($L6 \times C4$), as an institutional process unfolding ($L6 \times C2$), or, when the analysis turns to the operational cause, as a deliberate operation within the firm ($L4 \times C5$). None of these is the phenomenon's true type; each is a position taken for a purpose. The matrix positions; it does not sort.

The purpose of fixing these positions is verification. A language model emits propositions fluently but without an independent check on whether their structure is coherent. By assigning each proposition a cell, an external verifier can examine the trajectory of a chain of reasoning across the matrix and detect when that trajectory crosses non-adjacent positions without a bridging step — the failure this book calls Layer Jump. HNS-36 is therefore the structural substrate of Axis 2: the minimal grid on which structural incoherence becomes visible and attributable.

1.1 Reading HNS-36

Five principles govern every use of the matrix:

- Layers are causal. They answer where a phenomenon originates.
- Categories are explanatory. They answer how a phenomenon is being interpreted.
- Cells are positions, not contents. A cell defines a coordinate, not a prescribed meaning.
- Mapping is rule-based. Placement follows the procedures of Chapter 5, not intuition.

- The matrix is structurally neutral. No cell is higher or more important than another.

Note. *HNS-36 is the base of a nested family. Chapter 7 extends it to HNS-144 by adding the Onto-Modal Quadrants; Volume 3 extends that to HNS-864 by adding the Inference & Control Operators. The Social Meta Layers (Volume 4) are a parallel grounding axis (Axis 3) and are not part of this multiplicative nesting.*

1.2 Notation and conventions

Three conventions are used throughout. A cell is written $(L_i \times C_j)$ at HNS-36 resolution and $(L_i \times C_j \times Q_k)$ at HNS-144 resolution. A chain of reasoning is represented as a cell trajectory — the ordered sequence of cells assigned to its propositions — which the verifier inspects for non-adjacent transitions. Finally, sections marked normative state requirements; sections marked informative or non-normative supply motivation and examples and impose no requirement. Unless stated otherwise, the definitions and rules of this volume are normative.

2. The Two Axes

HNS-36 rests on two axes that must remain independent. The Natural Layers form the vertical axis (L) and describe causal origin. The Abstract Categories form the horizontal axis (C) and describe explanatory purpose. The cardinal error to avoid is treating categories as causal or layers as interpretive.

The two axes answer different questions. Of any proposition one may ask: “Where does this originate causally?” — a layer question — and “How am I interpreting it?” — a category question. A single phenomenon may legitimately be examined under different categories without changing its causal layer, and the same category applies across every layer. Keeping these roles distinct is the precondition for correct placement.

2.1 Why two independent axes multiply

The reason for two axes rather than one list is the same geometric intuition that motivates MAV. When two constraint axes are independent, the positions they jointly define are their product, and a proposition must satisfy both to be admissible. Independence is the decisive property: if the layer of a claim could be read off from its category, the second axis would add nothing. The specification therefore requires bidirectional independence — the meaning of a cell cannot be derived from either axis alone; it emerges only from their intersection.

The independence of the two axes is what makes their product informative rather than redundant. If layers and categories were correlated — if knowing that a claim is institutional told you it must be read as a process — the second axis would carry no information and the thirty-six positions would collapse toward six. The specification therefore treats orthogonality as a design target, not a discovered fact: the axes are chosen so that, so far as possible, any layer can be examined under any category. Where a particular cell is rare or strained in practice, that is information about the phenomenon, not a defect of the grid.

3. Human Natural Layers (L1–L6)

The Human Natural Layers are the six causal strata of the matrix, ordered from physical substrate to macro-social institution. They are the primary axis (L): every proposition is placed in exactly one layer before any category is applied. The canonical names and their conceptual lineage are as follows.

ID	Name	Lineage (inspiration)	Causal role
L1	Physical / Substrate	Physics; BFO Material Entity (Smith)	The material and biological substrate on which all higher layers depend.
L2	Cellular / Informational	Computational neuroscience	Neural representation: the encoding of signals into informational structure.
L3	Cognitive / System 1	Kahneman (Type 1)	Fast, intuitive processing; the generative, associative mode (maps to MAV Axis 1).
L4	Operational / System 2	Kahneman (Type 2)	Slow, deliberate, rule-governed verification (maps to MAV Axis 2/3).
L5	Micro-Social / Interaction	Habermas	Interpersonal interaction and communicative exchange; basis of SMS-2/3.
L6	Macro-Social / Institutional	Institutional theory; BFO Social Object	Institutions, norms, and macro-social systems; maps to SMS-5/6.

Note. Kahneman’s System 1 and System 2 are parallel dual processes, not an intrinsic vertical hierarchy. HNS re-positions them as processing-depth layers (L3, L4). This is a deliberate re-framing for the purposes of the coordinate, not a claim about Kahneman’s original model.

3.1 Layer structure requirements

Each layer must satisfy five conditions, which together make the axis well-formed:

- Causality — the layer represents a causal domain that produces effects in the layers above it.
- Non-redundancy — no layer duplicates the causal function of another.

- Completeness — the six layers jointly span the causal structure relevant to the framework (a chosen, not a proven, partition; see Chapter 12).
- Vertical ordering — layers are read from foundational (L1) to emergent (L6).
- Single assignment — a proposition is assigned to exactly one layer, by its primary causal origin and not by its effects, context, or scale.

3.2 Boundary conditions and misclassification

Ambiguous cases are resolved by origin, not appearance. A claim whose effects surface at a higher layer is still placed at the layer that produces it; a composite claim is decomposed into causal components, each placed separately. The recurring confusions to guard against are: reading a higher, emergent regularity (L6) as a cause; conflating the informational encoding of L2 with the cognitive processing of L3; and assigning a claim to the social layers (L5–L6) merely because its subject matter is social, when its causal origin is cognitive.

3.3 The six layers in detail

L1 — Physical / Substrate

L1 is the material and biological substrate on which every higher layer supervenes: anatomy, physiology, neural hardware, and the physical constraints of signal propagation and energy. It is causally first, in the precise sense that nothing in the layers above can occur without it. Its lineage is physics together with the BFO category of material entity (Smith), which supplies the formal notion of an independent material continuant.

Boundary and example. L1 is the physical substrate itself, not the information it carries: firing thresholds, metabolic limits, and propagation delays belong to L1, whereas the representation those signals encode belongs to L2. Non-normative example: a claim about a channel saturating above a fixed input rate is an L1 claim.

L2 — Cellular / Informational

L2 is the encoding of physical signals into informational, representational structure — the neural-representation layer. Its lineage is computational neuroscience, on the view that populations of cells carry information in their joint activity. Causally, L2 mediates between the physical substrate

below and cognition above: it is where matter becomes message.

Boundary and example. L2 is representation, not yet System-1 cognition. A population code, a place-cell map, or an embedding is L2; the fast inference that consumes them is L3. Example: a claim about how a quantity is encoded across a representational space is an L2 claim.

L3 — Cognitive / System 1

L3 is fast, automatic, associative processing: pattern completion, heuristic inference, the immediate impression. It is the generative mode and maps to MAV Axis 1. Its lineage is the Type-1 side of dual-process theory (Kahneman; Stanovich & West; Evans), re-positioned here as a processing-depth layer rather than a free-standing system.

Boundary and example. L3 is intuitive generation; L4 is deliberate checking. A snap judgement that a face is familiar is L3; verifying the identity against a record is L4. Example: a claim that an answer feels right is an L3 claim.

L4 — Operational / System 2

L4 is slow, deliberate, rule-governed processing: verification, planning, sequential reasoning. It maps to MAV Axis 2/3, the verification function. Its lineage is the Type-2 side of dual-process theory together with the integrative theory of prefrontal cognitive control (Miller & Cohen), in which the prefrontal cortex maintains goals and enforces rules against automatic responses.

Boundary and example. L4 differs from L3 as effortful from automatic. Multi-step arithmetic, or consciously applying a policy to a case, is L4. Example: a claim that a result was checked against an explicit rule is an L4 claim.

L5 — Micro-Social / Interaction

L5 is interpersonal interaction and communicative exchange: coordination, negotiation, the raising and redeeming of validity claims between agents. Its lineage is Habermas's theory of communicative action, in which meaning is secured through dialogue oriented to mutual understanding. L5 is the basis of the lower Social Meta Layers (SMS-2/3).

Boundary and example. L5 is event-level interaction between agents; L6 is the institutional regularity that governs such events. A specific negotiation is L5; the contract law that frames negotiations is L6.

Example: a claim about what two parties agreed in a meeting is an L5 claim.

L6 — Macro-Social / Institutional

L6 is institutions, norms, and macro-social systems together with their emergent regularities. Its lineage is institutional theory and the BFO category of social object (Smith), which gives the formal status of entities that exist by collective recognition. L6 maps to the upper Social Meta Layers (SMS-5/6).

Boundary and example. L6 is the macro or institutional structure, not a single interaction. Market price formation and regulatory regimes are L6; one transaction within them is L5. Example: a claim about how a market sets prices is an L6 claim.

4. Abstract Categories (C1–C6)

The Abstract Categories are the six non-causal explanatory axes of the matrix. Where the layers say where a proposition originates, the categories say how it is being interpreted. They form the horizontal axis (C); each applies across all six layers.

ID	Name	Lineage (inspiration)	Explanatory role
C1	Continuant / Entity	Aristotle; BFO	An object grasped as a persisting thing.
C2	Occurrent / Process	Aristotle; BFO	A temporally unfolding, event-like aspect.
C3	Relational / Boundary	Aristotle; Kant	Relations and boundaries between objects; aligns with the HNS orthogonality concept.
C4	Quality / State	Kant	A property or state at a given moment; the state-space view.
C5	Transformation / Operation	Pearl (do-operator)	A transition or operation between states; the core of causal transition.
C6	Meta-Level / Schema	Kant (categories of categories)	Reference to an object from a higher schematic level; aligns with the HNS runtime / EVA.

Note. Under strict BFO, Continuant (C1) and Occurrent (C2) are the top-level dichotomy, Quality (C4) is a dependent continuant, and Process overlaps with C5. The six categories here are therefore a synthetic set inspired by Aristotle, BFO, and Kant — not a flat partition derived from any single ontology. That six categories should tile representable structure descends philosophically from Aristotle’s Categories and Kant’s table of the categories, and computationally from upper ontologies such as DOLCE (Gangemi et al.), BFO (Smith), and SUMO (Niles & Pease) and from frame semantics (Fillmore).

4.1 Category structure requirements

- Non-causality — categories provide explanatory perspectives, never causal domains.
- Cross-layer applicability — each category applies to all six layers.

- Non-redundancy — no category duplicates another’s explanatory function.
- Completeness — the six jointly span the explanatory dimensions the framework requires (again, a chosen resolution).
- Single assignment — a proposition is interpreted through exactly one category at a time, by primary explanatory purpose and not by domain, scale, or importance.

4.2 The six categories in detail

C1 — Continuant / Entity

C1 interprets a phenomenon as a persisting thing — an object grasped as enduring through time. Its lineage is Aristotle’s notion of substance and the BFO continuant. Example: treating the institution as a standing entity is a C1 interpretation; its reform, as an unfolding event, would be C2.

C2 — Occurrent / Process

C2 interprets a phenomenon as a temporally unfolding event or process. Its lineage is Aristotle and the BFO occurrent. The boundary with C5 matters: a process (C2) is an unfolding occurrent, whereas a transformation or operation (C5) is the rule that maps one state to another. Example: a negotiation unfolding over a week is C2.

C3 — Relational / Boundary

C3 interprets a phenomenon through the relations and boundaries between objects, rather than the objects themselves. Its lineage is Aristotle’s category of relation and Kant; it aligns with the HNS notion of orthogonality. Example: the dependency between two departments, or the boundary separating two domains, is C3.

C4 — Quality / State

C4 interprets a phenomenon as a property or state at a given moment — the state-space view. Its lineage is Kant’s category of quality. The boundary with C2 is the snapshot/unfolding distinction: a state (C4) is a momentary configuration, a process (C2) is its change. Example: the system’s current load is C4; the load rising is C2.

C5 — Transformation / Operation

C5 interprets a phenomenon as a state transition or operation — the causal-intervention view. Its lineage is Pearl’s do-operator, which

distinguishes acting on a system from merely observing it. Example: applying a policy changes the outcome distribution — a C5 interpretation, an intervention rather than an unfolding.

C6 — Meta-Level / Schema

C6 interprets a phenomenon from a higher, schematic level — a category of categories, or a reference to the framework itself. Its lineage is Kant's schematism; it aligns with the HNS runtime and the External Verification Architecture. Example: the rule that governs which category applies to a claim is itself a C6 object.

5. The 36-Cell Specification

A cell is the intersection of one layer and one category, written $(L_i \times C_j)$. This chapter specifies the internal format of a cell and the rules that govern the matrix as a whole. It defines the form of a cell, not its content.

5.1 Cell format

Every cell is documented with the same ordered components: a cell identifier ($L_i \times C_j$); a layer reference (Chapter 3); a category reference (Chapter 4); a structural-role statement describing the function of the intersection; an interpretive scope giving the range of propositions it admits; boundary conditions distinguishing it from adjacent cells; misclassification-prevention guidance; and optional non-normative examples. This ordering is mandatory across all thirty-six cells.

5.2 Structural and matrix constraints

- Uniqueness — each cell is a unique structural position.
- Non-overlap — no two cells share the same structural domain.
- Bidirectional independence — a cell's meaning cannot be derived from its layer or category alone.
- Interpretive neutrality — cells define positions, not prescribed content.
- Exhaustiveness and mutual exclusivity — all layer $\times$ category intersections are represented, and a single decomposed proposition occupies one cell at a time.

5.3 Adjacency and interpretation

Cells are adjacent in two ways only: same layer, neighbouring category (horizontal), or same category, neighbouring layer (vertical). Adjacency does not imply similarity; boundary conditions distinguish neighbours. Interpretation proceeds layer-first (the causal domain fixes the primary meaning), then category-second (the explanatory purpose refines it). Two interpretive prohibitions hold: no cross-cell inference — a cell's meaning may not be read off from another — and no diagonal interpretation — a simultaneous step in both layer and category has no defined structural meaning, and is precisely the signature of a Layer Jump (Chapter 6).

5.4 Mapping rules

To place a proposition: (1) identify the unit of analysis; (2) determine its primary causal domain and assign one layer; (3) determine its primary explanatory purpose and assign one category; (4) combine them into ($L_i \times C_j$); (5) verify the boundary conditions of Chapters 3 and 4; (6) record the placement in the notation of the template (Appendix B). Composite phenomena are decomposed before mapping; once mapped, a placement is stable unless the explanatory purpose changes.

5.5 Representative cell entries

The following non-normative entries illustrate the template in use. They document four representative cells; the full set of thirty-six follows the same form (Appendix B).

Cell ($L4 \times C5$) — Operational $\times$ Transformation

- Structural Role: positions claims about deliberate operations that transform a state under operational control.
- Interpretive Scope: planned interventions and applied procedures whose effect is the object of analysis.
- Boundary Conditions: distinguished from ($L4 \times C2$), the unfolding of a process, and from ($L3 \times C5$), an automatic rather than deliberate transformation.
- Misclassification Prevention: do not place a merely associative, automatic transition here; that belongs to L3.
- *Example (non-normative)*: running the verification routine attenuates ungrounded output.

Cell ($L6 \times C1$) — Macro-Social $\times$ Continuant

- Structural Role: positions claims about institutions grasped as persisting entities.
- Interpretive Scope: organizations, legal entities, and markets considered as standing objects.
- Boundary Conditions: distinguished from ($L6 \times C2$), an institution undergoing reform, and from ($L5 \times C1$), a single relationship treated as an entity.

- Misclassification Prevention: do not assign here on grounds of scale alone; macro subject matter is not automatically macro-social in causal origin.
- *Example (non-normative)*: the regulator is the body responsible for enforcement.

Cell (L3 × C4) — Cognitive (System 1) × Quality / State

- Structural Role: positions claims about momentary intuitive states.
- Interpretive Scope: a felt impression or a fast confidence estimate at an instant.
- Boundary Conditions: distinguished from (L4 × C4), a deliberate assessment, and from (L3 × C2), the unfolding of an intuition over time.
- Misclassification Prevention: do not treat the report of a feeling as an objective measurement; that crosses into the objective quadrants at HNS-144.
- *Example (non-normative)*: the answer feels right.

Cell (L2 × C2) — Cellular / Informational × Occurrent

- Structural Role: positions claims about informational processes unfolding within neural representation.
- Interpretive Scope: encoding dynamics and the propagation of activation across a representational space.
- Boundary Conditions: distinguished from (L1 × C2), a physical process, and from (L2 × C5), an operation performed on the representation.
- Misclassification Prevention: do not elevate a representational process to cognition (L3); encoding is not yet inference.
- *Example (non-normative)*: activation spreads across the embedding as context accumulates.

Cell (L1 × C3) — Physical × Relational / Boundary

- Structural Role: positions claims about physical couplings and boundaries between material entities.
- Interpretive Scope: interfaces, contact, and physical dependency between substrates.

- Boundary Conditions: distinguished from ($L1 \times C1$), the entities themselves, and from ($L3 \times C3$), an associative cognitive link.
- Misclassification Prevention: a conceptual or metaphorical relation is not a physical one; keeping them apart guards against Metaphor Contamination.
- *Example (non-normative)*: the two components are thermally coupled.

Cell ($L5 \times C5$) — Micro-Social $\times$ Transformation / Operation

- Structural Role: positions claims about communicative acts that change an interpersonal state.
- Interpretive Scope: speech acts and moves in a negotiation that transform commitments.
- Boundary Conditions: distinguished from ($L5 \times C2$), the dialogue unfolding, and from ($L4 \times C5$), a non-social operation.
- Misclassification Prevention: do not treat a unilateral cognitive operation as a communicative act.
- *Example (non-normative)*: the offer, once accepted, binds both parties.

Cell ($L4 \times C6$) — Operational $\times$ Meta-Level / Schema

- Structural Role: positions claims about explicit rules or meta-policies that govern operations.
- Interpretive Scope: policies, verification rules, and the schema by which operations are selected.
- Boundary Conditions: distinguished from ($L4 \times C5$), the operation itself, and from ($L6 \times C6$), an institutional norm.
- Misclassification Prevention: a meta-rule is not the same as the act it governs.
- *Example (non-normative)*: the policy specifies which checks run before output.

Cell ($L6 \times C2$) — Macro-Social $\times$ Occurrent / Process

- Structural Role: positions claims about institutional processes and macro trends unfolding over time.
- Interpretive Scope: reforms, market movements, and long-run institutional change.

- Boundary Conditions: distinguished from ($L6 \times C1$), the institution as entity, and from ($L5 \times C2$), a single interaction unfolding.
- Misclassification Prevention: do not read an emergent macro trend as an individual cause — a frequent source of Layer Jump.
- *Example (non-normative)*: the market has been consolidating over the decade.

Cell ($L1 \times C1$) — Physical $\times$ Continuant / Entity

- Structural Role: positions claims about material bodies grasped as persisting objects.
- Interpretive Scope: organs, devices, and physical substrates treated as things.
- Boundary Conditions: distinguished from ($L1 \times C2$), a physical process, and from ($L2 \times C1$), a representational unit.
- Misclassification Prevention: do not confuse the material body with the information it carries.
- *Example (non-normative)*: the server is the physical host of the service.

Cell ($L2 \times C5$) — Cellular / Informational $\times$ Transformation

- Structural Role: positions claims about operations that transform a representation.
- Interpretive Scope: re-encoding, projection, or compression of a code.
- Boundary Conditions: distinguished from ($L2 \times C2$), the encoding process unfolding, and from ($L4 \times C5$), a deliberate operation.
- Misclassification Prevention: an automatic re-encoding is not a deliberate operational intervention.
- *Example (non-normative)*: the vector is projected into a lower-dimensional space.

Cell ($L3 \times C2$) — Cognitive (System 1) $\times$ Occurrent / Process

- Structural Role: positions claims about automatic inference unfolding over time.
- Interpretive Scope: a train of associations, a spreading activation of intuition.

- Boundary Conditions: distinguished from (L3 × C4), a momentary felt state, and from (L4 × C2), controlled reasoning.
- Misclassification Prevention: do not treat an intuitive cascade as deliberate reasoning.
- *Example (non-normative)*: one association triggered the next without reflection.

Cell (L4 × C4) — Operational × Quality / State

- Structural Role: positions claims about a deliberately assessed state.
- Interpretive Scope: a measured or judged condition arrived at by reasoning.
- Boundary Conditions: distinguished from (L3 × C4), a fast felt state, and from (L4 × C2), the assessment process.
- Misclassification Prevention: do not record an intuitive impression as a deliberate assessment.
- *Example (non-normative)*: after analysis, the project is judged behind schedule.

Cell (L5 × C3) — Micro-Social × Relational / Boundary

- Structural Role: positions claims about interpersonal relations and their boundaries.
- Interpretive Scope: trust, authority, and role relations between agents.
- Boundary Conditions: distinguished from (L5 × C1), a party as an entity, and from (L6 × C3), an inter-institutional relation.
- Misclassification Prevention: a personal relation is not an institutional one.
- *Example (non-normative)*: the two teams share a reporting line.

Cell (L6 × C6) — Macro-Social × Meta-Level / Schema

- Structural Role: positions claims about norms and institutional schemas that govern macro behaviour.
- Interpretive Scope: legal principles, standards, and the constitutive rules of institutions.
- Boundary Conditions: distinguished from (L6 × C5), a specific regulation, and from (L4 × C6), an individual meta-policy.

- Misclassification Prevention: a society-wide norm is not a single actor's rule.
- *Example (non-normative)*: the principle of equal treatment governs the institution.

5.6 The thirty-six cells at a glance

The grid below gives a concise structural tag for every cell. It is a map, not a substitute for the full entries above. Rows are Natural Layers (L1–L6); columns are Abstract Categories (C1–C6).

	C1 Entity	C2 Process	C3 Relation	C4 State	C5 Operation	C6 Schema
L1	object	process	coupling	state	transfer	law schema
L2	code unit	encoding	code map	code state	recoding	code schema
L3	percept	intuition	association	felt state	heuristic	frame
L4	plan step	reasoning	rule link	judgement	operation	policy
L5	party	dialogue	role relation	rapport	speech act	protocol
L6	institution	macro trend	macro link	macro state	regulation	norm

6. Layer Jump

Layer Jump is the failure HNS-36 is built to detect: the traversal of non-adjacent representational layers without an explicit bridging proposition. In the matrix it appears as a cell trajectory that crosses non-adjacent rows with no transitional step — the structural signature of an argument that has skipped the causal chain connecting its premises to its conclusion. In clinical terms it is the analogue of formal thought disorder; in language models it is the confident leap from one level of explanation to a distant one.

HNS-36 screens for it geometrically. Each proposition in a chain receives a cell; the verifier inspects successive cells for non-adjacent transitions. A jump from a macro-social cell (L6) to a cognitive cell (L3) with no intervening operational or interactional step is flagged, and the offending step is attenuated rather than emitted. The check is structural, not statistical: it does not ask whether the leap is improbable, only whether it is bridged.

Worked example (non-normative)

Prompt: *“Why did the company’s stock fall?”*

Failing output: *“The stock fell because the employees were unmotivated.”*

Diagnosis: a jump from the macro-social/price layer to the individual-cognitive layer with no transitional proposition — the cell trajectory crosses non-adjacent rows.

Corrected: *“The stock fell on weak quarterly earnings (price layer). Earnings were weak partly because output declined (operational layer); internal surveys link lower output to reduced motivation (individual layer). The bridge runs motivation → productivity → earnings → valuation.”*

A second case. Prompt: *“Why did the patient recover?”* Failing output: *“Because they had a positive attitude.”* This jumps from an individual-cognitive claim to a physiological outcome with no bridging mechanism. Corrected: *“Recovery followed the prescribed course of treatment (physiological); adherence to it was high (operational); a positive attitude may have supported adherence (individual). The supported bridge runs attitude → adherence → treatment effect → recovery.”*

Formally, the check is on the trajectory. Each proposition receives a cell;

the verifier reads the sequence and marks any step whose layers are non-adjacent and unbridged. The test is local and mechanical — it asks only whether consecutive cells are neighbours, or are joined by an explicit transitional proposition — which is what makes a Layer Jump detectable without recourse to the plausibility of the content.

Part II
The Observational Extension — HNS-144

7. HNS-144: The Observational Extension

HNS-144 raises the resolution of the matrix. Each of the thirty-six cells is subdivided by the Onto-Modal Quadrants — four quadrants formed by the Cartesian product of two binary dimensions — giving $36 \times 4 = 144$ cells. The two dimensions are the Ontological Domain (Subjective / Objective) and the Modal Invariants (Necessary / Contingent).

Dimension	Binary values	Lineage	Role
① Ontological Domain	Subjective / Objective	Kant; Habermas	Separates how something is held from how something is.
② Modal Invariants	Necessary / Contingent	Kant (modality)	Distinguishes a rule from a context-dependent fact.

Their product yields the four quadrants:

Quadrant	Definition
Q1	Subjective $\times$ Contingent
Q2	Subjective $\times$ Necessary
Q3	Objective $\times$ Contingent
Q4	Objective $\times$ Necessary

***Note.** The resolution is two dimensions, not four. Two binary dimensions give $2 \times 2 = 4$ quadrants, hence $36 \times 4 = 144$. Four binary dimensions would give $2^4 = 16$ sub-states and $36 \times 16 = 576$, which is not the specified cardinality. Earlier informal descriptions that listed four “logical dimensions” conflated the number of axes with the number of resulting quadrants; the corrected and canonical reading is two dimensions yielding four quadrants.*

7.1 The four quadrants in detail

Q1 — Subjective $\times$ Contingent

Q1 holds claims about how something is held rather than how it is, and that could have been otherwise: preferences, impressions, reports of feeling. Example: the team found the meeting useful.

Q2 — Subjective $\times$ Necessary

Q2 holds subjective claims asserted as binding within a frame — commitments, declared values, the constitutive rules of a practice. Example: in our charter, a quorum is required to vote.

Q3 — Objective × Contingent

Q3 holds matters of fact that could have been otherwise: measurements, observations, empirical outcomes. Most empirical claims belong here. Example: the model's accuracy was 0.91 on this test set.

Q4 — Objective × Necessary

Q4 holds objective claims that hold of necessity — definitions, mathematical truths, logical entailments. Example: in Euclidean geometry the interior angles of a triangle sum to 180 degrees.

Most Category Ambiguity failures are quadrant errors: a Q1 preference dressed as a Q4 necessity, or a Q3 contingency asserted as Q4. The observational layer's task is to hold each claim to a single declared quadrant, or to require its decomposition into several.

8. Category Ambiguity

Category Ambiguity is the failure HNS-144 is built to observe: the conflation of distinct categories — most commonly the collapse of the subjective/objective distinction and of the fact/value (normative) distinction into a single undifferentiated claim. Where Layer Jump is a fault of causal traversal, Category Ambiguity is a fault of conceptual boundary: a sentence that treats a preference, an outcome, and a value judgement as though they were one kind of thing.

HNS-144 observes it by enforcing concept-boundary conditions at quadrant resolution. A proposition that ought to occupy the subjective-contingent quadrant but is asserted as if objective-necessary is flagged for boundary violation; the verifier requires the claim to be re-stated within a single, declared quadrant or explicitly decomposed into several.

Worked example (non-normative)

Prompt: *“Is the new policy good?”*

Failing output: *“Yes, it’s good because everyone likes it.”*

Diagnosis: a subjective preference (‘is liked’), an objective outcome (‘achieves its goal’), and a normative judgement (‘is good’) are collapsed — the subjective/objective distinction is crossed and a contingent preference is presented as a necessary conclusion.

Corrected: “Three distinct questions. (1) Is it liked? Survey data say yes — subjective, contingent. (2) Does it achieve its stated goal? Early metrics are mixed — objective, contingent. (3) Is it good, all things considered? A normative judgement depending on which values are weighted. Popularity bears on (1), not directly on (3).”

A second case. Prompt: “Is this design better?” Failing output: “Yes, it’s more modern, so it’s better.” Here an aesthetic-subjective property is asserted as an objective-necessary superiority, crossing both the subjective/objective and the contingent/necessary boundaries. Corrected: “It is more recent in style (objective, contingent). Whether ‘modern’ is preferred is a subjective judgement (subjective, contingent). Whether it is better depends on a stated criterion — usability, cost, accessibility — each measurable separately.”

9. Mapping and Usage at 144 Resolution

At full resolution a proposition is placed at $(L_i \times C_j \times Q_k)$: a layer, a category, and a quadrant. The mapping procedure of Chapter 5 extends directly — after layer and category are fixed, the verifier determines the ontological domain and modal status and selects the quadrant. The same constraints apply: single assignment per decomposed unit, stability of placement, and decomposition of composite claims before mapping.

The 144 resolution is observational in the precise sense that it is where category conflations become visible that the 36 resolution cannot see. Two propositions may share a cell at HNS-36 yet differ in quadrant — one a contingent observation, the other asserted as a necessary rule — and it is exactly this difference that the observational layer is designed to surface. HNS-144 does not introduce new layers or categories; it refines the existing thirty-six positions to the resolution at which boundary violations can be detected and attributed.

Worked mapping at 144 resolution (non-normative)

Take the claim “interest rates must rise because inflation is high.” The layer is macro-social and institutional — a claim about monetary policy (L6). The category is a transformation: a policy operation said to follow from a condition (C5). The quadrant is fixed by two words: “must” asserts necessity, and the claim is framed as objective, placing it at Q4 (Objective $\times$ Necessary). The full coordinate is therefore $(L6 \times C5 \times Q4)$.

The observational layer now does its work. It asks whether the asserted necessity is warranted. Rates often rise with inflation, but not invariably; the relationship is in fact contingent. The claim has thus been placed at Q4 when it belongs at Q3 (Objective $\times$ Contingent), and the quadrant mismatch is flagged as a modal overreach — a necessity claimed where only a tendency holds. The correction restates it as contingent. This is precisely the conflation HNS-144 is built to observe and that HNS-36 alone, lacking the modal dimension, cannot see.

Composite claims

A real sentence often carries several propositions at once, and these may fall in different cells and quadrants. The rule is to decompose before mapping: split the sentence into its atomic claims, place each, and only then assess the trajectory across them. “The popular reform will obviously raise growth” decomposes into a subjective-contingent claim

about popularity (Q1), an objective-contingent prediction about growth (Q3), and a modal overreach in the word “obviously.” Mapping the parts separately is what lets the verifier accept the defensible components while flagging the overreach, rather than passing or rejecting the sentence whole.

Part III

Integration and Status

10. Relationship to the HNS Architecture

HNS-36 and HNS-144 are the lower two resolutions of Axis 2, the Structural / Causal axis of MAV. They sit within a larger architecture, and their boundaries with the neighbouring specifications should be stated plainly.

10.1 The nested coordinate (Axis 2)

Specification	Composition	Cells
HNS-36	Natural Layers (6) × Abstract Categories (6)	36
HNS-144	HNS-36 × Onto-Modal Quadrants (4)	144
HNS-864	HNS-144 × Inference & Control Operators (6)	864

HNS-864 (the subject of Volume 3) adds the six Inference & Control Operators — 01–03 from Pearl’s hierarchy of association, intervention, and counterfactual, and 04–06 from Powers’ perceptual control theory plus an HNS output regulator — to reach 864 atomic-proposition cells. HNS-36/144 supplies the coordinate on which those operators act.

10.2 The orthogonal grounding axis (Axis 3)

The Social Meta Layers (HNS-SMS-6, Volume 4) constitute Axis 3, the Grounded / Executive axis. They are orthogonal to the nesting above: they ground and regulate output against context, from the somatic base to the universal, but they do not subdivide the coordinate and are not a multiplicative factor. The complete normative reference for all of these axes is reproduced in Appendix A.

10.3 The runtime

In deployment, the checks specified here are applied by a pre-decode interception layer that assigns cells to candidate propositions and attenuates those whose trajectory violates the adjacency and boundary conditions. The runtime is described in the Origin Trilogy and specified at operator resolution in Volume 3; for HNS-36/144 it suffices that each correction is attributable to an explicit cell, which is the source of the framework’s auditability.

10.4 Relationship to the three MAV axes

HNS-36/144 specifies Axis 2, but it helps to state how the three Multi-

Axis Verification axes divide the labour. Axis 1 (Generative / Statistical) is the model's native fluency — the System-1 layer (L3) that proposes continuations. Axis 2 (Structural / Causal), specified here, supplies the layer-and-category constraints a proposal must satisfy, independent of how probable it is. Axis 3 (Grounded / Executive), the Social Meta Layers of Volume 4, checks the proposal against context from the body to the institution. Reliability, on the MAV account, is the conjunction: a claim must be fluent, structurally coherent, and grounded. HNS-36/144 owns the middle condition and is silent on the other two by design.

This division also says what the coordinate does not do. It does not judge whether a claim is true — an Axis-3 grounding question against the world — nor whether it is probable, an Axis-1 question. It judges whether the claim's structure is coherent: whether its layers are bridged and its categories kept distinct. A claim can be perfectly structured and still false; HNS-36/144 will pass it, and Axis 3 must catch it. Keeping these responsibilities separate is what makes each axis auditable.

11. Intellectual Lineage and Proponents

This chapter consolidates the lineage of HNS-36/144 and names the proponents of the ideas it draws upon. As stated at the outset, these are inspirations, not derivations; HNS recombines them into a coordinate for external verification.

11.1 The layers

The causal-control reading of the upper layers descends from the integrative theory of prefrontal cortex (Miller & Cohen) and the hippocampal account of episodic and schematic binding (Eichenbaum). The L3/L4 split realizes the dual-process distinction (Stanovich & West; Evans & Stanovich; Kahneman), re-positioned as processing depth. The micro-social layer (L5) draws on Habermas's theory of communicative action; the macro-social layer (L6) on institutional theory and on the BFO category of social object (Smith). The physical base (L1) is anchored in physics and in BFO's material entity.

11.2 The categories

The six categories descend from Aristotle's *Categories* and Kant's table of the categories of understanding, and computationally from the upper ontologies — DOLCE's *endurant/perdurant/quality/abstract* distinctions (Gangemi et al.), BFO's *continuant/occurrent* split (Smith), and SUMO (Niles & Pease) — and from frame semantics (Fillmore). The transformation category (C5) takes its admissibility standard from Pearl's account of causal intervention; the binding of features into coherent objects without illusory conjunction is the binding problem (Treisman & Gelade; von der Malsburg).

11.3 The quadrants

The Ontological Domain and Modal Invariants dimensions are Kantian in origin — the subjective/objective distinction and the category of modality (necessity/contingency) — with the communicative reading of the subjective/objective boundary drawing on Habermas.

The proponents named here did not design HNS, and several would frame their work differently. Pearl's hierarchy is a theory of causal inference, not of language-model verification; Kahneman's dual-process account is descriptive psychology, not a layered architecture; BFO is an ontology for

scientific-data integration. HNS borrows their distinctions and recombines them for an engineering purpose. The borrowing is acknowledged precisely so that the additions stay visible: the coordinate idea, the output-attenuation operator, and the use of the matrix as an external verifier are HNS's own.

The deeper debt is to the tradition of upper ontologies itself — the conviction, running from Aristotle through Kant to BFO, DOLCE, and SUMO, that what can be said admits a small, principled set of top-level distinctions. HNS-36 is a wager that such a set, fixed in advance and made machine-checkable, is useful for constraining generated text. Whether the wager pays is the empirical question of the next chapter.

11.4 Lineage at a glance

HNS element	Drawn from	Proponent(s)
Layers L1-L2	Physics; computational neuroscience; upper ontology	Smith (BFO)
Layers L3-L4	Dual-process theory; prefrontal control	Kahneman; Stanovich & West; Miller & Cohen
Layers L5-L6	Communicative action; institutional theory	Habermas; Smith (BFO)
Categories	Categories; upper ontology; frames; causal inference	Aristotle; Kant; Gangemi; Smith; Fillmore; Pearl
Quadrants	Subjective/objective; modality	Kant; Habermas
Coordinate idea	Geodetic framing of description	HNS (this volume)

12. Empirical Status, Limitations, Threats to Validity

A specification is only as strong as its honesty about what it has established. HNS-36/144 makes a small number of secure claims and a larger number of proposals and conjectures, and they should not be confused.

12.1 What is established

- Well-definedness. The coordinate is fully specified: two axes, thirty-six cells, four quadrants, and the rules for placement. Given the rules, placements are reproducible.
- Auditability by construction. Because every correction is attributed to an explicit cell, the matrix yields a machine-checkable record of how an output was constrained. This is true by construction, independent of whether the partition is the right one.

12.2 What is proposed

- That the layer×category coordinate captures the structure relevant to verification, and that Layer Jump and Category Ambiguity are the failure classes it most naturally screens. These are engineering proposals, to be tested against external criteria rather than framework-internal ones.

12.3 What is conjecture

- Completeness of the partition — that six layers and six categories tile the representable structure — is a hypothesis, to be tested by peer review and adversarial probing as upper ontologies are tested.
- The correspondence between cells and identifiable brain functions is a structured analogy, not a proven mapping.

12.4 On the number six

The recurrence of six — six layers, six categories, and six operators in the higher specification — is a chosen resolution, not a natural constant. No source ontology has exactly six categories; Aristotle lists ten and Kant twelve, while BFO's fundamental division is twofold. The layer stack likewise reflects a deliberate partition in which dual-process and

micro/macro distinctions are each resolved into two. The specification is stronger for stating this openly: six is adopted for coverage, memorability, and architectural consistency, and the framework does not claim that exactly six is metaphysically necessary. Where a reviewer presses on why six, the honest answer is that it is the resolution the programme has selected, and that the partition is open to revision under the standard of proof set out above.

12.5 Threats to validity

Two threats deserve emphasis. First, framework-internal scoring: a coordinate designed to detect Layer Jump will, unsurprisingly, detect Layer Jump, so any reported suppression is weaker than a reduction measured against an external criterion such as factual accuracy. Second, the partition mixes organizing principles — the layers blend a scale ordering with a processing-depth distinction — and a stricter formulation might separate these. Both are reasons to treat HNS-36/144 as a useful, auditable scaffold whose completeness remains an open question, not as a finished metaphysics.

A further limitation deserves naming. Because a proposition's placement depends on an interpreter's reading of its primary causal origin and explanatory purpose, two careful analysts may occasionally disagree on a cell. The specification reduces but does not eliminate this: the mapping rules, boundary conditions, and adjacency constraints narrow the choices, and a disagreement itself is informative, localizing exactly where an argument is ambiguous. Inter-rater reliability of cell assignment is therefore a concrete, measurable target for validation — and a more honest figure of merit than any framework-internal suppression count.

Two boundaries also limit scope. The line between Axis 2 and Axis 3 is not always sharp: whether a claim's defect is structural — a category crossed — or a matter of grounding — a fact wrong about the world — can itself be a judgement, and the coordinate can flag only the former. And the partition has been developed chiefly against English-language, general-domain reasoning; whether the same six layers and six categories transfer cleanly across languages and specialist domains is untested. Both are reasons to treat the present specification as a stable but revisable v1.0, not a closed result.

Appendix A (Normative) — HNS Unified Structural Coordinate Table v2.0

The following is the canonical reference for every HNS structural primitive. The lineage columns indicate conceptual inspiration, not strict derivation from a single ontology.

A.1 Natural Layers (HNS-36)

ID	Name	Lineage (inspiration)
L1	Physical / Substrate	Physics / BFO Material Entity
L2	Cellular / Informational	Computational neuroscience
L3	Cognitive / System 1	Kahneman Type 1
L4	Operational / System 2	Kahneman Type 2
L5	Micro-Social / Interaction	Habermas
L6	Macro-Social / Institutional	Institutional theory / BFO Social Object

A.2 Abstract Categories (HNS-36)

ID	Name	Lineage (inspiration)
C1	Continuant / Entity	Aristotle / BFO
C2	Occurrent / Process	Aristotle / BFO
C3	Relational / Boundary	Aristotle / Kant
C4	Quality / State	Kant
C5	Transformation / Operation	Pearl (do-operator)
C6	Meta-Level / Schema	Kant (category of categories)

A.3 Onto-Modal Quadrants (HNS-144)

Quadrant	Definition
Q1	Subjective × Contingent
Q2	Subjective × Necessary
Q3	Objective × Contingent
Q4	Objective × Necessary

A.4 Inference & Control Operators (HNS-864)

ID	Name	Proposer
01	Conditioning	Statistics / Pearl
02	Intervention / do-operator	Pearl
03	Counterfactual	Pearl
04	Reference Signal	Powers (PCT)
05	Comparator / Error Detection	Powers (PCT)
06	Output Attenuation	HNS Engineering

A.5 Social Meta Layers (HNS-SMS-6) — parallel grounding axis (Axis 3)

ID	Domain	Lineage (inspiration)
SMS-1	Somatic	Barsalou / Harnad
SMS-2	Protocol	Habermas
SMS-3	Organizational	HNS Extension
SMS-4	Economic	HNS Extension
SMS-5	Governance	Institutional theory
SMS-6	Universal	Ashby / General Systems Theory

Note. Composition: $HNS-36 = 6 \times 6$; $HNS-144 = 36 \times 4$; $HNS-864 = 144 \times 6$.
 The Social Meta Layers are orthogonal to this nesting and are not multiplied into it.

Appendix B — The 36-Cell Template (Formal)

Each of the thirty-six cells is documented with the following template. The template defines structure, not content.

Cell ($L_i \times C_j$)

- Layer: name of the Human Natural Layer (L_i), per Chapter 3.
- Category: name of the Abstract Category (C_j), per Chapter 4.
- 1. Structural Role — the function of this layer×category intersection.
- 2. Interpretive Scope — the range of propositions the cell admits.
- 3. Boundary Conditions — rules distinguishing the cell from its horizontal and vertical neighbours.
- 4. Misclassification Prevention — the common errors to avoid.
- 5. Non-normative Examples (optional) — illustrations of the cell's structural meaning.

Appendix C — Worked Examples

C.1 Layer Jump (HNS-36)

Prompt: *“Why are sales up this quarter?”*

Failing output: *“Sales are up because we changed the logo.”*

Diagnosis: a leap from a macro-social/market outcome to a single surface change, skipping the operational layer that would connect them.

Corrected: “Sales rose 12%. The spring promotion was A/B-tested and measured a +8% uplift (operational); seasonal demand follows the historical pattern (market); the rebrand was coincident with no measured effect. Only the promotion has a supported bridge to the outcome.”

C.2 Category Ambiguity (HNS-144)

Prompt: *“Was the meeting productive?”*

Failing output: *“Yes — it felt productive and that’s what matters.”*

Diagnosis: a subjective-contingent impression (‘felt productive’) is asserted as an objective-necessary conclusion (‘that’s what matters’); two quadrants are collapsed.

Corrected: “Subjectively, participants reported it felt productive (subjective, contingent). Objectively, two of three agenda items were decided and one was deferred (objective, contingent). Whether that counts as productive depends on the standard applied.”

Glossary

HNS-36 — The base coordinate: a 6×6 matrix of Natural Layers $\times$ Abstract Categories; the Structural OS.

HNS-144 — The observational extension: HNS-36 $\times$ the four Onto-Modal Quadrants; the Observational OS.

Natural Layers (L1-L6) — The six causal strata forming the vertical axis; from physical substrate to macro-social institution.

Abstract Categories (C1-C6) — The six non-causal explanatory axes forming the horizontal axis.

Onto-Modal Quadrants (Q1-Q4) — Two binary dimensions — Subjective/Objective and Necessary/Contingent — whose product yields four quadrants.

Cell ($L_i \times C_j$) — A structural position: the intersection of one layer and one category.

Layer Jump — Traversal across non-adjacent layers without a bridging proposition; screened by HNS-36.

Category Ambiguity — Conflation of distinct categories or quadrants; observed by HNS-144.

Bidirectional independence — The property that a cell's meaning cannot be derived from its layer or category alone.

Auditability — The property that every correction is attributable to an explicit cell and logged.

MAV (Multi-Axis Verification) — The account that reliability arises from the intersection of three approximately orthogonal verification axes.

Axis 2 (Structural / Causal) — The verification axis specified by HNS-36/144: constraints on structure, category, and causal order, independent of frequency.

External Verification Architecture (EVA) — The runtime architecture (Volume 5) in which the coordinate checks are applied to model output.

Output Attenuation — The operator that scales an output by a grounding score, so ungrounded content is suppressed rather than emitted.

Grounding function $G(x)$ — The Axis-3 function that returns 1 only when an output is compatible with all six Social Meta Layers.

Onto-Modal — Pertaining to the two HNS-144 dimensions: ontological domain (subjective/objective) and modality (necessary/contingent).

Cell trajectory — The ordered sequence of cells assigned to the propositions in a chain of reasoning, inspected for non-adjacent transitions.

Structured analogy — A correspondence used for design intuition without a claim of exact identity, as in the brain-region mappings.

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